Leong Wai Men

Game & Engine Programmer

Seeking a Game Client / Engine Programming Internship

Email: wai.men.leong@gmail.com
Portfolio: waimenleong.wixsite.com/website
LinkedIn: linkedin.com/in/leong-wai-men
GitHub: github.com/waimen-leong
Games: Google Drive game builds

Game & engine programmer studying Computer Science (Real-Time Interactive Simulation) at DigiPen Institute of Technology Singapore, graduating September 2028. Tech Lead and Engine Champion on Umapyoi Engine, a custom C++ engine our team of 7 used to ship Senki, a 2D hack-and-slash.

Skills

Languages: C++, C#, C, Lua

Engine & graphics: OpenGL, ImGui, Sol2, Unity

Online & data: PlayFab, Photon (PUN2), MySQL

Tools: Git / GitHub, Trello, Android Studio (Java)

Team communication & conflict management

Projects

Umapyoi Engine & Senki / Custom C++ Game Engine
TECH LEAD & ENGINE CHAMPION · DIGIPEN · TEAM OF 7 · [DATES]

- Led the engine's technical direction for a 7-person team; the engine was used to ship Senki, a 2D top-down hack-and-slash set in mythological feudal Japan.
- Built the ECS core (entity manager, components, systems) that the engine and gameplay run on.
- Built the prefab system, handling edge cases where scene data overwrote prefab data and nested prefabs broke each other's dependencies.
- Built the Lua scripting system (Sol2), collision / physics and controller input.
- Built the tilemap system with an in-engine editor, editor undo / redo, and a PlayFab leaderboard.
- Implemented enemy AI behaviour for Senki.
- Source: github.com/Siestasies/UmapyoiEngine

Death Dungeon / 2D Multiplayer Action Mobile Game
OCTOBER 2021 - JANUARY 2022, SINGAPORE

- Fast-paced arena fighting game for Android, built solo in Unity.
- Online multiplayer with a self-developed lag compensation feature.

- Player login, including Google / Facebook login, with player data stored in a locally hosted MySQL database.

Escape From Dungeon / 2D Local Multiplayer Platformer
OCTOBER 2021 - JANUARY 2022, SINGAPORE

- Year 2 studio team project built in Unity for Windows and WebGL.
- Built local 2-player multiplayer on a single computer.
- Built enemy AI using a state machine.
- Built the audio manager and game UI (main menu, pause screen).

Experience

Metascape City / Game Developer
MAY 2022 - AUGUST 2022, SINGAPORE

- Worked on the early stage of a multiplayer WebGL app where users explore a virtual recreation of Singapore and interact with each other.
- Added 3D buildings to the environment and optimised 3D model rendering performance using BVH-based techniques.
- Implemented minor multiplayer interactions between users.

VizioFly / Software Developer (Internship)
MARCH 2021 - AUGUST 2021, SINGAPORE

- Built interactive Unity applications for clients, including a Virtual Expo where users walk around an open area and interact with booths.
- Built an Online Presentation feature for hosting presentations on a WebGL website.
- Implemented Photon (PUN2) multiplayer, UI, NavMesh navigation and Facebook integration in Unity.
- Built server-side features in PHP, HTML and JavaScript, using MySQL and PlayFab with Facebook login for user data and authentication.

Ritz-Carlton Hotel / Banquet Server (Part-time)
DURING SCHOOL HOLIDAYS, SINGAPORE

- Served guests as part of a banquet team at dinner events, and led small groups to complete assigned tasks.

Education

DigiPen Institute of Technology Singapore & Singapore Institute of Technology / BSc Computer Science in Real-Time Interactive Simulation
AUGUST 2024 - SEPTEMBER 2028 (EXPECTED), SINGAPORE

- Relevant coursework: Data Structures & Algorithms, Computer Networks, Computer Graphics, Linear Algebra, Discrete Mathematics

Nanyang Polytechnic / Diploma in Game Development & Technology
MARCH 2019 - MARCH 2022, SINGAPORE

- Distinction in Studio Project (Game Project)
- Coursework: OpenGL / C++ graphics, physics, server & database, multiplayer and console game development, level design